

New Research Study on Exploratory Testing

Reference paper: An experiment on the effectiveness and efficiency of exploratory testing — Afzal, Ghazi, Itkonen, Torkar, Andrews, Bhatti, *Empirical Software Engineering* 20(2):844–878, 2015. **DOI:** 10.1007/s10664-014-9301-4

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Proposed title: *Does AI-assisted Exploratory Testing change the effectiveness/efficiency trade-off? A replication of Afzal et al. (2015) with LLM-supported testers in higher education.*

Goal (Goal–Question–Metric)

Analyse the practice of exploratory testing **for the purpose of** evaluating the impact of an LLM-based testing assistant (a chat model with charter context but no test-case generation) on tester productivity **with respect to** defect-detection efficiency, defect-detection effectiveness (difficulty / type / severity), and false-defect rate **from the point of view of** the tester and the testing researcher **in the context of** MSc Software Verification & Validation students performing time-boxed manual functional testing on a medium-size open-source application (jEdit, replicating Afzal et al.'s SUT).

Research Questions

- **RQ1 (efficiency).** Do testers performing ET with an LLM assistant find more defects in a fixed 90-minute session than testers performing classic ET without an LLM?
- **RQ2 (effectiveness).** Do AI-assisted ET sessions differ from unassisted ET sessions in the *difficulty*, *type* and *severity* of the detected defects?
- **RQ3 (cost of assistance).** Does the LLM assistant increase the number of *false defect reports* compared to unassisted ET?
- **RQ4 (process).** How does the assistant change the testing *flow* — proportion of time spent on charter reading, exploration, oracle reasoning, note-taking?

Methodology

Controlled experiment with one factor (testing approach) and two treatments — ET–plain and ET–LLM — replicating Afzal et al.'s design. ~40 top-performing MSc students, randomised into balanced groups, each runs one 90-minute session on jEdit 4.2 seeded with the same 25 faults as in the original paper, preceded by a 15-minute briefing and a common charter. The ET–LLM group has access to a chat interface backed by a fixed model (e.g. Claude Sonnet, temperature 0) with the user-manual chapters and the charter pre-loaded into the system prompt; chat logs are recorded. Both groups submit a defect report and a session log; defects are independently triaged for type (IEEE 1044 categories), severity (1–3), and detection difficulty (4 modes). A short post-session survey collects perceived helpfulness, trust, and friction. Statistical analysis follows the reference paper: Mann–Whitney U at $\alpha=0.05$ for between-group comparisons, Vargha–Delaney $\hat{A}_{12}$ for effect size, and Wilcoxon for repeated-measures sub-analyses.

Metrics

Construct	Metric	Collection
Efficiency	# true defects per 90-min session	Defect-report triage
Efficiency (process)	Defects per active testing minute (excluding chat)	Session log + chat log
Effectiveness — difficulty	Defects per detection-difficulty mode (1–4)	Triage on Afzal et al.'s scheme
Effectiveness — type	Defects per IEEE 1044 type	Triage
Effectiveness — severity	Defects per severity level (1–3)	Triage
False-defect cost	# invalid defect reports / total reports	Triage
AI-interaction load	Chat turns; tokens in/out; % of session in chat	Chat-tool telemetry
Tester perception	Likert scores: helpfulness, trust, friction	Post-session survey
Confounders	Years of dev/testing experience, prior LLM usage	Pre-session demographic form